Seen, Then Deleted: 4D Occupancy World Models Erase the Agents They Cannot Forecast

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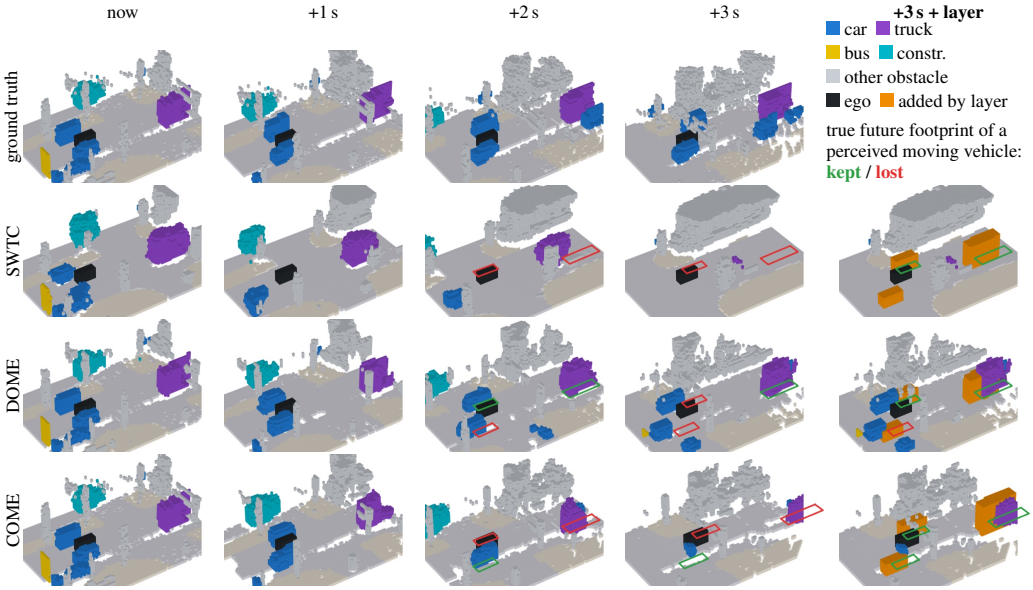


Figure 1. **Seen, then deleted: one held-out frame, three world models**, each row the model's own 3D occupancy output (0.4 m voxels; SparseWorld-TC (SWTC) from cameras, DOME and COME from the true occupancy up to *now*). All three perceive the car overtaking on the ego's left at 8.7 m/s; by +3 s none has it on its true footprint (red outline beside the ego): SWTC has turned the road into static structure, DOME leaves the car behind, COME drops it. The permanence layer (last column) keeps it for all three (green). It also shows the layer's limit: DOME loses the car on the right, and the detector's mispredicted motion puts the layer's box (orange) behind it, so it stays red. Outlines and kept/lost follow the paper's metric ($\geq$50% of the footprint at $p \geq 0.2$); the frame was chosen as one where all three models lose agents. A failure case of the layer is in Fig. 5.

Abstract

Driving world models are increasingly used to judge *actions: a planner's candidate paths are rolled out in an imagined 4D future and vetoed if they collide. We test a minimal requirement for such a judge, that an agent it has just perceived does not disappear from the future it imagines. Three pretrained 4D occupancy world models from different families (camera-based trajectory-conditioned, latent diffusion, deterministic forecasting) fail it: on held-out nuScenes scenes, 59–79% of the moving vehicles they perceive are no longer covered by their imagined occupancy 3 s later, including vehicles that remain clearly visible. The deleted agents are the ones the model cannot forecast: for them, even an oracle choice among the model's own predicted blobs is placed correctly for only 12–17%, against 35–40% for the motion predicted by a camera detector. Measuring this needs care: probability-weighted and any-cell perma-*

nence metrics are won by a blank uniform map or by scattered dots, so we propose a footprint-coverage versus false-area frontier that resists both. A training-free permanence layer that keeps every perceived agent as an object reduces deleted agents by 13.5, 15.5 and 6.6 points at equal false area in a pre-registered held-out test on all three models. For decisions, a physics safety check on the same detections, given the same at-fault rule, remains as good or better except at the rarest veto rates: permanence is necessary, not sufficient. Every main result survived two from-scratch reimplementations that import no analysis code, which also overturned several of our own earlier claims.

1. Introduction

World models, and the world action models built on them, are increasingly asked to judge actions: imagine each candidate trajectory's future and veto the unsafe ones [1, 9, 22]. A verdict is only as trustworthy as the imagined world, so a world model used as a test oracle must itself be validated [13], and voxel or pixel metrics do not say whether an imagined future is usable for action [10, 25]. We test one minimal requirement, *object permanence*: an agent the model has just perceived must not vanish from the future it imagines (Fig. 1). The COME authors show DOME suddenly dropping objects in qualitative examples [16], vanishing dynamic objects were reported for convolutional occupancy-grid predictors [7, 8, 19, 20], transporting occupancy instead of regenerating it is a growing design principle [12, 22, 23], and permanence is open for video world models [24]; to our knowledge it has not been measured for pretrained 4D occupancy world models, the forecasting setup introduced by OccWorld [26]. BeyondSight [14] keeps occluded actors in end-to-end perception and shows that SparseDrive, our detector, drops them under occlusion; we instead test whether world models keep agents that remain visible. **Contributions.**

- *Diagnosis:* three model families delete most perceived moving vehicles within 3 s; it is neither occlusion nor the latent representation, and their benchmarks [11, 18] do not measure it.
- *Association:* the deleted agents are those the model cannot forecast; we read this as per-cell occupancy turning "unsure where" into "nothing there", an association and not a measured mechanism (Sec. 4).
- *Measurement:* two natural permanence metrics are gameable; a footprint-coverage versus false-area frontier resists the blank-map and dot-lattice controls we tried, and is what an occupancy collision check needs.
- *Reference fix and decisions:* a training-free permanence layer is a reference that learned fixes must beat; it restores agents on all three models (pre-registered), and for decisions a fair physics check stays as good or better.

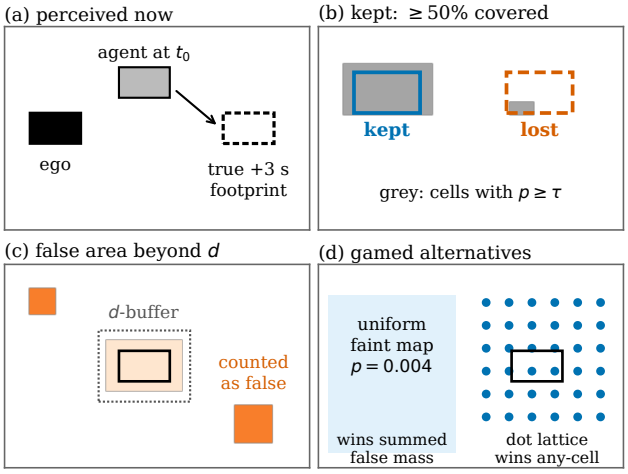


Figure 2. **The permanence measure** (Sec. 2, Table 2). (a) A moving vehicle perceived now and its true +3 s footprint. (b) It is kept if ≥50% of the footprint has $p \geq \tau$ (blue, solid), else lost (vermillion, dashed). (c) False area counts cells with $p \geq \tau$ farther than d from every true vehicle. (d) The alternatives are gamed: a uniform faint map wins summed false mass, a dot lattice wins the any-cell criterion. (Figs. 1, 5, 6 show kept/lost in green/red.)

Every confirmatory test was committed before its held-out outputs were computed; two from-scratch reimplementations that import no analysis code checked every number (Sec. 9.6).

2. Setup and a permanence measure that resists simple gaming

World models. SparseWorld-TC (SWTC [5]) imagines 200×200×16 occupancy (0.4 m) at +1/2/3 s from six cameras, conditioned on the ego path; our wrapper of the released SWTC checkpoint gives lower forecasting accuracy than published (mIoU 20.9 vs. 25.6 averaged over 1–3 s on 600 val frames), so all SWTC statements are within-model comparisons; we did not diagnose the shortfall, and the 59.3% cannot be tied to the published model. DOME [6] (latent diffusion) and the scene-centric forecasting branch of COME [16] (deterministic UNet) receive four ground-truth Occ3D frames [18], so what they lose is lost by prediction alone. All COME results are for this forecasting branch, not the full COME system. DOME is run with one diffusion sample per window (seed variability is not measured and the intervals do not include sampling variance). Collision checks are top-down: we project each voxel column to its maximum vehicle probability over car height, which can only make a car look *more* present. **Data and protocol.** nuScenes val, split by source scene into folds A (75) and B (74); Adv-nuSc [21] adds 156 attack scenes for decisions, where Impromptu-VLA [2] proposes 16 trajectories per keyframe. Everything is developed on fold A;

each confirmatory test (Sec. 9.7) was committed with code hashes before its held-out outputs were computed. **Measure.** Pre-registration IDs (H1–H4, A1–A13) are defined in Table 3. Each model defines its own perceived cohort, so the three headline rates are on different vehicle sets (sizes in Table 1). A vehicle moving >1 m/s within 40 m is *perceived* if the model's current map has mean vehicle probability $\geq$0.3 in its box; it is *kept* at threshold τ if $\geq$50% of its true +3 s footprint has $p \geq \tau$, else *lost*. False area(τ) counts cells with $p \geq \tau$ farther than d from any true vehicle. Methods are compared by lost at the frozen model's false area at $\tau = 0.2$ (scene-cluster bootstrap); both axes are invariant to any monotone recalibration. **Why not simpler measures** (Fig. 2d). Summed probability ("false mass") rewards spreading: a blank map with $p = 0.004$ everywhere loses *no* car at frozen SWTC's false mass. "Kept if any cell in the box is above τ" rewards sparse dots: a dot lattice around moving detections beats frozen SWTC by 19 points. Under footprint coverage the dots are 15 points *worse*; blurring and a rigid "nothing moves" carry gain nothing at $d = 0.4$ m (Sec. 9.2). The layer's test was registered at $d = 3$ m (Table 1); $d = 0.4$ m closes a smaller loophole and gives equal or larger gains (Sec. 9.3).

3. Diagnosis: seen, then deleted

On held-out scenes the frozen models lose 59.3% (SWTC), 79.4% (DOME) and 71.4% (COME) of the moving vehicles they perceive by +3 s (Table 1; fold A 59 / 78 / 75%), 50–72% already at +2 s, and 54–81% across stricter perception, faster or nearer agents and Occ3D's own vehicle voxels as truth (Sec. 9.3). Even under the lenient any-cell criterion, 25–51% are lost. **Not occlusion:** among vehicles still clearly visible at +3 s (82–88% of them), 57–77% are lost. **Not the representation:** passing the *true* future through DOME's own VAE loses 4.6% of perceived vehicles (0.1% any-cell). **Two ways to delete.** SWTC keeps its total vehicle probability constant but lets it drift off the agents; DOME and COME lose it outright, 20% and 35% of it by +3 s (Table 1, mass).

4. Why: the deleted agents are the ones the model cannot forecast

If the model knew where a deleted car went but failed to keep it, its forecast should still hold the location. It does not, even when we give it every advantage: among all blobs of its +3 s forecast within the agent's reachable disc we pick the one *closest to the truth*. For deleted agents this oracle blob lies in the true box for only 12–17% on fold B (9–13% on fold A; Table 4), whereas the camera detector's motion prediction places 35–40% (Fig. 4a). For kept agents, DOME's and COME's forecasts beat the detector (81–87% vs. 58–65%); SWTC's does not (43% vs. 58%). The models delete exactly the agents whose future they cannot predict. We read this as per-cell occupancy having no notion of an agent, so a car that may be in many places becomes faint probability everywhere and, at any useful threshold, nothing anywhere. **This is an association, not a proof of mechanism.** The deleted/kept split and the oracle test use the same forecast, so part of the link is built in; we did not measure probability dispersion directly, and the detector also predicts deleted agents worse than kept ones (35–40% vs. 58–65%), so forecasting difficulty may explain part of it. The VAE control rules out the latent representation for DOME only.

5. Fix: never let a perceived agent vanish

Since the deleted agents are those the world model cannot place, the fix must not rely on it to place them. The **permanence layer** keeps every moving vehicle that the camera detector [17] reports now (score $\geq$0.3, $\geq$1 m/s; nuScenes-train checkpoint, run online), places it at the detector's most likely motion at each horizon, renders its box into the imagined future and takes the cellwise maximum with the world model's map. It is training-free and needs no access to the model's internals. **Pre-registered test (A11, fold B, once):** it lowers lost vehicles at equal false area by 13.5, 15.5 and 6.6 points (Table 1, Fig. 3); supported on all three models. **Both parts are needed:** the detected boxes alone lose 73% of SWTC's perceived vehicles, more than the world model. **Robust:** a re-implementation from the raw detector output matches to 0.1 points; the gain holds at $d = 0.4$ m, at +2 s (-18 / -20 / -9), at twice the false-area budget (at half of it for SWTC; for DOME and COME the layer cannot reach so little false area), for every population variant and under the any-cell criterion (Sec. 9.3); it fails where the detector misses an agent or mispredicts its motion (Fig. 5). **Learned alternatives.** A transport readout trained on fold A (fusing the forecast with the transported current map) also helps (-14.9 / -13.2 / -3.9) and needs the forecast; fine-tuning SWTC on Occ3D targets helps too (-11 to -17 on held-out drives) but retrains the world model and worsened its ranking of dangers.

6. Decisions: physics is a strong baseline

Safety envelopes such as RSS [15] have been proposed as a layer under the planner, a rule-based planner beat learned ones [3], yet world-model verifiers are compared only with learned methods [9]. We compare with physics (constant velocity on the same detections) at matched false-veto rates on real safe paths, with the label's at-fault rule (NAVSIM-style [4]: ignore contact while stopped or from behind) applied to *both* checkers. At 1–5% of safe paths vetoed, physics catches 57–80% of real at-fault collisions and 23–41% of attacker collisions, the world model 51–78% and

Table 1. **Headline results** (held-out fold B, +3 s, $\tau = 0.2$). Lost: share of perceived moving vehicles whose true footprint is <50% covered; any-cell: lenient criterion; visible: still ≥60% visible with ≥5 lidar points; mass: total imagined vehicle probability relative to the true vehicle area (SWTC: constant total, but 10% less near true vehicles). Layer: pre-registered A11 at the frozen model's false area ($d = 3$ m as registered); Bonferroni-adjusted 95% CI; the stricter $d = 0.4$ m column is from Sec. 9.3. Perceived moving vehicles: $n = 3{,}077$ (SWTC) and 3,502–3,503 (DOME, COME).

model (input)	lost	any-cell	visible	mass $t_0 \to +3$ s	+ layer	Δ [CI], d=3 m	Δ, d=0.4 m
SWTC (cameras)	59.3%	24.5%	56.8%	1.57→1.58	45.8%	−13.5 [−17.1, −10.2]	−13.5
DOME (GT occupancy)	79.4%	50.9%	77.0%	0.93→0.75	63.9%	−15.5 [−19.6, −11.8]	−15.7
COME (GT occupancy)	71.4%	41.3%	69.8%	0.94→0.61	64.8%	−6.6 [−12.7, −0.3]	−7.9

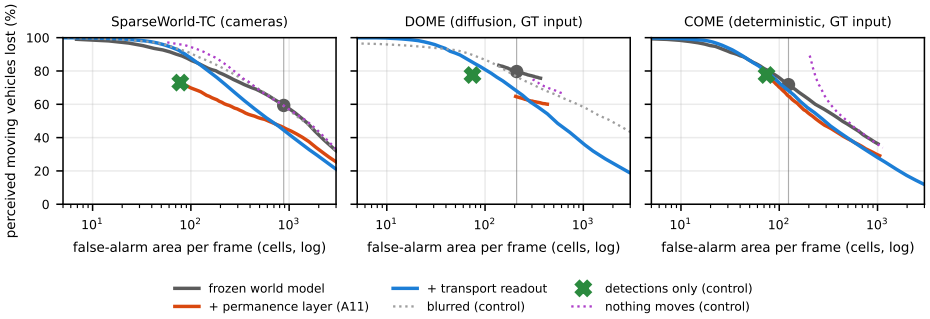


Figure 3. **Held-out frontiers** (fold B, +3 s, $d = 0.4$ m): perceived moving vehicles lost vs. false-alarm area as the threshold sweeps. Dot and line: the frozen model at $\tau = 0.2$. The permanence layer (orange) lies below each frozen frontier at its operating point; the detected boxes alone (green cross) and trivial controls do not.

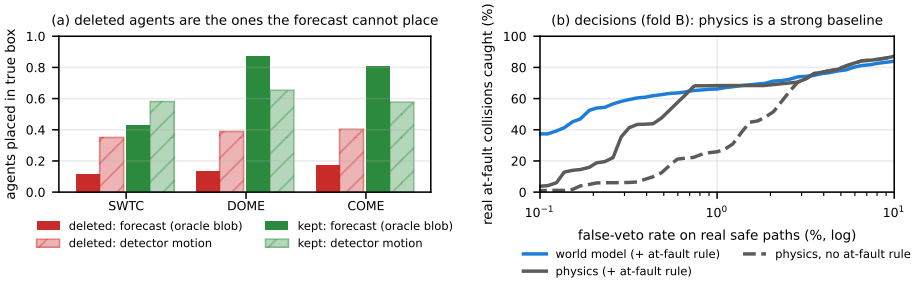


Figure 4. **(a) Cause** (fold B): agents placed inside their true box at +3 s by the forecast's best blob (oracle choice, solid) and by the detector's motion prediction (hatched), for deleted and kept agents. **(b) Decisions** (fold B, real at-fault collisions): recall vs. false-veto rate. With the label's at-fault rule on both checkers, physics matches the world model from about 1% of safe paths vetoed upward; without the rule (dashed) physics looks far weaker than it is.

16–33% (both folds); adding the permanence layer to the verifier does not change this (A12, not supported). The world model leads only when vetoes are very rare (Fig. 4b): at 0.25–0.5% it catches more real collisions (fold A 31–40 vs. 25–39%, fold B 56–62 vs. 21–49%), not attacker ones. Our pre-registered A13 found a large gain from giving the world model part of a 1% budget (+32 points) but was registered against physics *without* the at-fault rule; with the rule on both sides the gain disappears. The detector misses the object that is hit in 9–30% of real and 33–54% of attacker collisions, where occupancy is the only safeguard, and 84% of the attackers that SWTC deleted end >2 m from their constant-velocity path: a change of intent that no checker foresees from the current state.

7. Limitations

Open loop, one dataset, one attack generator, 3 s horizons, vehicles only (no pedestrians or cyclists); two of three models receive ground-truth occupancy, and the three cohorts differ, so we do not claim the rates are comparable across models. Covering half of a vehicle's footprint is not the same as predicting occupancy along a candidate ego path, and a drop in coverage can also reflect displacement or shape error, not only disappearance. The detector (SparseDrive) is the only one tried, so the layer, the cause comparison and the physics baseline all share its errors; we did not run the layer with ground-truth boxes. The decision results (Fig. 4b) are reported without counts or scene-

Table 2. Change in lost vs. frozen SWTC for informative and uninformative maps.

change in lost vs. frozen SWTC (fold B, points)	any-cell, d=0.4 m	coverage, d=3 m	coverage, d=0.4 m
dot lattice around moving detections (no information)	-19.3	n/a*	$+14.8$
Gaussian blur σ=1.6 m (no information)	$+11.3$	-6.8	-0.4
max-filter dilation 2 m (no information)	$+12.4$	-5.1	0.0
rigid carry of the current map ("nothing moves")	-1.0	$+0.2$	-0.3
permanence layer	-10.5	-13.5	-13.5
transport readout	-2.1	-15.7	-14.9

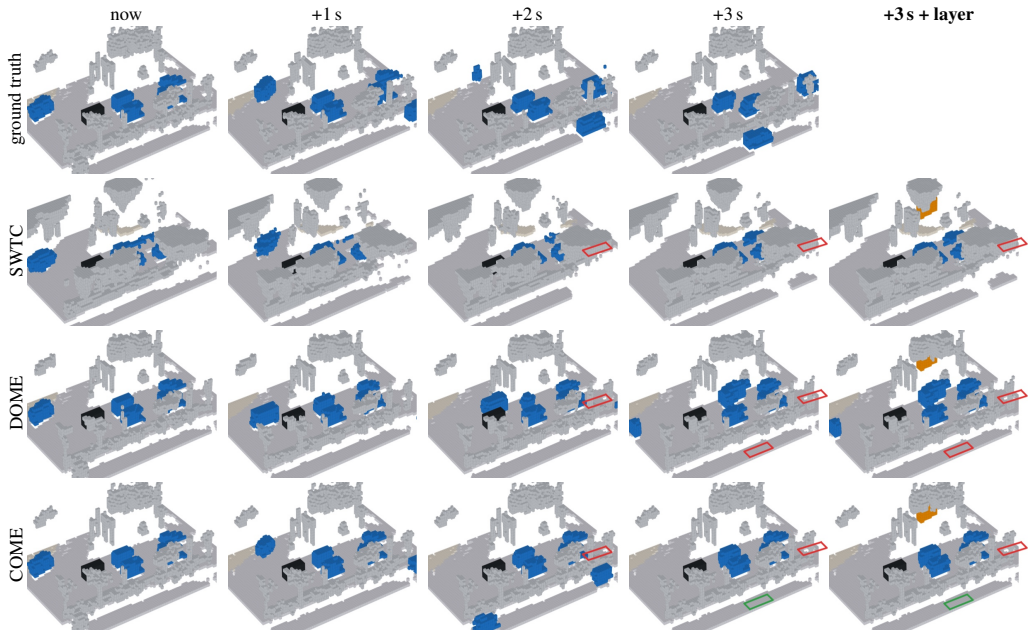


Figure 5. **A failure case** (held-out scene-0794; the ego is stopped). Lost / restored: SWTC 4 of 4 / 0, DOME 9 of 9 / 0, COME 7 of 9 / 0. Of DOME's nine, the detector misses five (no detection within 4.8 m), reports one slow car (1.4 m/s) as stationary, so the layer, which keeps moving detections only, drops it, and places the other three off their true footprints. The layer can only keep what perception provides.

Table 3. **Every registered test** (A6 was never registered), committed before its held-out outputs were computed (fold B or registered fallback). †at 2.6× false vehicle mass. ‡endpoint on summed false mass, later shown to be gameable (Sec. 9.6); superseded by A11. §supported as registered; does not hold with the at-fault rule on both checkers (Sec. 6).

Test	Endpoint	Estimate [95% CI]	Verdict
H1	adversary vs. matched ordinary recall	-0.06 $[-0.23, 0.05]$	not supported
H2	lateral motion → recall (per m)	-0.72 $[-0.86, -0.58]$	supported
H3	permanence + conformal, adv. recall	$+0.05$ $[-0.07, 0.17]$	not supported
H4	adversarial vs. masked fine-tuning	–	not run (stop rule)
A1	permanence guard, Adv collision	-0.12 pts $[-0.56, 0.33]$	not supported
A2	early-stopped readout, attacker recall	$+0.04$	not supported
A3	×10 loss weight, lost vs. control	-0.038 $[-0.051, -0.027]$	supported†
A4	SWTC∪readout, real AUPRC vs. CV	$+0.30$ $[+0.18, +0.40]$	supported
A5	relabelled loss, lost vs. control	-0.026 $[-0.035, -0.018]$	supported
A7	relabelled loss, Adv collision	-0.5 pts $[-1.2, 0.0]$	non-inferior
A8	readout on DOME, lost at equal false mass	-0.259 $[-0.323, -0.205]$	superseded (gameable endpoint, Sec. 9.6)‡
A9	swap (train B, test A): SWTC / DOME	-0.045 / -0.249	superseded (gameable endpoint, Sec. 9.6)‡
A10	map-aware readout, lateral lost	-0.016 / -0.066	not supported
A11	**permanence layer**, lost at equal false area: SWTC	-0.135 $[-0.171, -0.102]$	**supported**
	DOME	-0.155 $[-0.196, -0.118]$	**supported**
	COME	-0.066 $[-0.127, -0.003]$	**supported**
A12	layer in the verifier, AUPRC real / Adv vs. SWTC	-0.223 / $+0.010$	not supported
A13	split veto at 1% false vetoes, real recall vs. physics	$+0.320$ $[+0.108, +0.428]$	superseded§
	attacker recall vs. physics	$+0.054$ $[-0.073, +0.183]$	not supported

Table 4. Share of agents placed in their true box at +3 s.

		deleted agents		kept agents	
model	fold	oracle forecast blob	detector motion	oracle forecast blob	detector motion
SWTC	A / B	0.094 / 0.116	0.371 / 0.351	0.349 / 0.425	0.513 / 0.581
DOME	A / B	0.126 / 0.135	0.403 / 0.389	0.825 / 0.870	0.540 / 0.654
COME	A / B	0.134 / 0.174	0.391 / 0.404	0.760 / 0.809	0.548 / 0.578

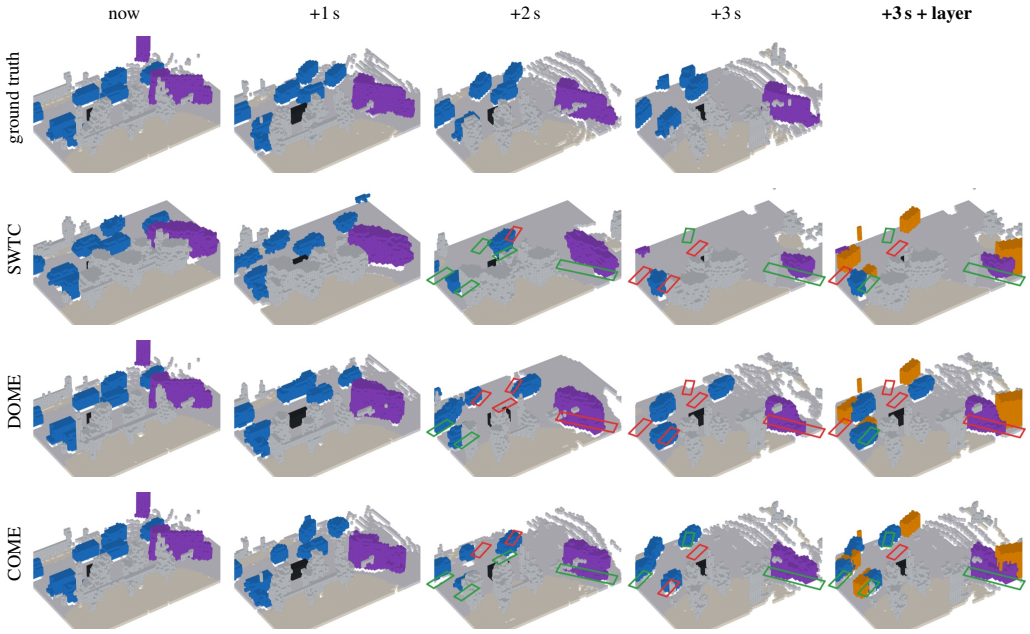


Figure 6. **Dense traffic** (held-out scene-0564; legend as in Fig. 1). Lost at +3 s / restored by the layer: SWTC 6 of 10 / 3, DOME 8 of 10 / 3, COME 5 of 10 / 3.

level intervals here, and we do not identify which interactions (e.g. occluded cut-ins, multi-agent yielding) benefit uniquely from an occupancy model; the physics check matches the world model only at veto rates of about 1% and above. The layer relies on an external detector: agents it misses or mis-predicts stay deleted, and it does not foresee intent. A permanence-aware detector such as BeyondSight [14] would directly reduce the layer's misses (Fig. 5). The measure is top-down and counts vehicles only, and the world model's veto thresholds transfer poorly between scene sets. We recommend reporting permanence as footprint coverage against false area, comparing verifiers with a physics check under the same at-fault rule, and carrying perceived agents as objects, not only as cells.

8. Conclusion

Pretrained 4D occupancy world models delete most of the moving vehicles they have just perceived, and the deleted ones are exactly those they cannot forecast. Footprint coverage against false area measures this without being gamed by blank maps or dots; a training-free permanence layer restores agents on all three models, while a physics check on the same detections remains as good or better for decisions. In a cascaded world action model the imagined future is the action critic, so deletion is a failure of exactly that component: a car that may be in many places becomes faint probability everywhere and, at any useful threshold, nothing anywhere.

9. Additional analyses

9.1. Setup details

Reproducibility. All analyses run on public data (nuScenes, Occ3D, Adv-nuSc) and official checkpoints; every confirmatory test is a git commit with SHA-256 hashes of its code (Sec. 9.7); both audits are self-contained scripts that import no analysis code. Code, registration files and audit logs will be released with the paper. Experiments, analysis code and both re-implementation audits were carried out with an LLM coding agent (Claude Code) under author direction; the authors checked and take responsibility for all results and text. **Models.** SWTC: official

checkpoint, conditioned on the logged ego path (for decisions: on each candidate); vehicle probability = softmax summed over Occ3D vehicle classes (bus, car, construction vehicle, trailer, truck), maximum over the height band -0.5 to 2.0 m. DOME: official checkpoint, four ground-truth Occ3D frames, one diffusion sample per window (seeded); our wrapper reproduces its published forecasting accuracy (mIoU/IoU averaged over 1–3 s on 24 scenes: 0.28/0.37; published 0.27/0.36). COME: forecasting branch (`unet_past2s_future3s`), softmax kept instead of argmax. **Population.** Vehicles except (motor)cycles, speed >1 m/s (nuScenes box velocity), within 40 m at t_0, true +3 s centre within 38 m of the ego; perceived if the frozen model's mean vehicle probability in the t_0 box is $\geq$0.3. Fold B: 3,077 (SWTC) and 3,502–3,503 (DOME, COME) vehicles in 64–68 scenes. **Frontier comparison.** For each method, lost(τ) and mean false area(τ) over a grid of 55 thresholds; lost is interpolated at the frozen model's false area at $\tau = 0.2$; the bootstrap resamples scenes and rematches the threshold in every resample (B = 1,000; amendment 11 uses Bonferroni over the three models). **Layer geometry.** Detected box length and width, oriented along the detected velocity, at the highest-scoring of six motion modes at +1/2/3 s; transformed into the ego frame of each moment with the logged ego poses, as the world models.

9.2. Gameable and robust permanence measures

In Table 2, *marks a method whose false area never falls to the reference. Under probability-weighted false mass, a uniform map with $p = 0.0039$ (SWTC) or 0.0021 (DOME) loses 0% at the frozen model's false mass, and rescaling a model's probabilities moves its frontier arbitrarily; area-based axes are exactly invariant to such rescaling (verified: $\times$0.01 leaves lost unchanged). The 3 m zone within which cells are never counted as false lets spreading gain up to 7 points; $d = 0.4$ m removes this.

9.3. Robustness of the diagnosis and the fix

Fold B, change in lost at equal false area (d=0.4 m) for the permanence layer, SWTC / DOME / COME: base $-13.5/-15.7/-7.9$; perception $\geq$0.5: $-13.6/-15.5/-7.7$; speed $>$3 m/s: $-14.0/-17.2/-11.6$; within 25 m: $-13.7/-18.8/-10.1$; +2 s: $-17.7/-20.2/-9.1$; half the false-area budget: -16.8 / n.a. / n.a. (the layer cannot go below its own boxes); twice the budget: $-9.4/-15.5/-9.4$; any-cell criterion: $-10.5/-20.4/-14.1$. Fold A gives the same picture ($-14.2/-14.6/-8.9$). Frozen lost at $\tau = 0.2$ under truth variants (SWTC / DOME / COME, fold B): $\geq$30% of the footprint 50.5 / 69.2 / 60.7%; $\geq$70% 70.9 / 89.3 / 83.4%; Occ3D vehicle voxels as footprint 59.2 / 78.6 / 71.1%.

9.4. Cause: forecast vs. physics on deleted and kept agents

Share of agents whose estimated +3 s centre lies inside the true box grown by 0.4 m. Forecast blobs: 8-connected components with $p \geq 0.1$ whose centroid lies within the reachable disc (speed$\times$3 s + 4 m) around the agent's current position; the oracle picks the blob closest to the truth. Detector motion: the matched detection's highest-scoring mode.

9.5. Qualitative results and further analysis

All 3D panels are the models' own outputs regenerated for these frames and identical to the evaluated ones (SWTC on the logged path and DOME with the evaluated batch and seed reproduce the stored top-down maps with IoU 1.000; COME is deterministic). Voxels show each model's most likely class; outlines and kept/lost follow the paper's metric (vehicle probability $\geq$0.2 on $\geq$50% of the true footprint), which is more lenient, so a footprint can count as kept where the most likely class is not a vehicle. Counts refer to all perceived moving vehicles within 40 m, not only those inside the crop. Legend as in Fig. 1; a dense-traffic example is in Fig. 6.

9.6. Audits and corrections

Audit 1 (a from-scratch re-implementation that imports no analysis code) reproduced every number of our earlier draft on its own axis and found that the axis, summed false mass, is won by a blank uniform map; on area, the readout's gain on SWTC vanished under the any-cell criterion (-2.3 $[-5.0, +0.1]$ at $d = 3$ m; -2.1 at 0.4 m) and shrank by two thirds on DOME. It also found two results attributed to the readout that belonged to its union with SWTC, an inconsistent ablation number, and a wrong claim ("DOME needs no forecast"). We retracted the earlier claims (fixed and summed-mass frontiers, "the model knows where the agent goes", mass conservation as evidence for the readout), derived the footprint/area measure, developed the permanence layer on fold A and registered A11 and A12. **Audit 2** rebuilt every result from raw nuScenes, Occ3D, model outputs and detector files without importing earlier code, rediscovered all grid conventions and horizon offsets, verified that detections carry no future information (0.78 m from the truth at t vs. 4.5 m at $t\pm$0.5 s), and confirmed Secs. 3–6; the 3D outputs shown were regenerated and match the evaluated ones (top-down IoU 1.000).

9.7. Pre-registration log

Commit hashes for the registration and every amendment, and the deviations (one pre-output bug fix in A8, documented), will be released with the code. A3/A5/A7 concern loss-level fine-tuning of SWTC on the earlier gate measure; re-evaluated on the footprint measure, the fine-tunes keep

more agents, without an advantage of the permanence losses over relabelled targets.

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